CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 5: Quadratic Equations

Topic-wise short notes with solved examples

A quadratic equation has the variable squared. This chapter covers solving it by factorisation and by the formula, giving answers to a required accuracy, and finding the nature of its roots.

1 Standard form and roots

- A **quadratic equation** in x has the form

$$ax^2 + bx + c = 0, \quad a \neq 0$$

- a, b and c are real numbers. If $a = 0$ the equation is no longer quadratic.
- A value of x that satisfies the equation is a **root** (solution). A quadratic equation has two roots, which may be equal.
- To check whether a number is a root, substitute it and see whether both sides become equal.

Example 1.1 If $x = 2$ is a root of $3x^2 - px - 2 = 0$, find p.

SOLUTION

Substitute $x = 2$: $3(4) - 2p - 2 = 0$

$10 - 2p = 0$

$p = 5$

2 Solving by factorisation

- Write the equation in the form $ax^2 + bx + c = 0$, clearing fractions and brackets first.
- Split bx into two terms whose sum is b and whose product is ac, factorise, and put each factor equal to zero (**zero product rule**: if $pq = 0$ then $p = 0$ or $q = 0$).

Example 2.1 Solve $2x^2 - 7x + 6 = 0$.

SOLUTION

$ac = 12$; the numbers -4 and -3 have sum -7 and product 12.

$2x^2 - 4x - 3x + 6 = 0$, so $2x(x - 2) - 3(x - 2) = 0$ and $(2x - 3)(x - 2) = 0$

$x = \frac{3}{2}$ or $x = 2$

Example 2.2 Solve $x + \frac{1}{x} = 2\frac{1}{2}$, $x \neq 0$.

SOLUTION

$x + \frac{1}{x} = \frac{5}{2}$. Multiply by $2x$: $2x^2 + 2 = 5x$, so $2x^2 - 5x + 2 = 0$

$(2x - 1)(x - 2) = 0$

$x = \frac{1}{2}$ **or** $x = 2$

Example 2.3 Solve $\frac{x}{x-1} + \frac{x-1}{x} = 2\frac{1}{2}$.

SOLUTION

Let $\frac{x}{x-1} = y$. Then $y + \frac{1}{y} = \frac{5}{2}$, which gives $y = 2$ or $y = \frac{1}{2}$ (as in the previous example).

$\frac{x}{x-1} = 2$ gives $x = 2x - 2$, so $x = 2$

$\frac{x}{x-1} = \frac{1}{2}$ gives $2x = x - 1$, so $x = -1$

$x = 2$ **or** $x = -1$

3 The quadratic formula

- The roots of $ax^2 + bx + c = 0$ are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Use the formula when the equation does not factorise, and always when the question says "correct to ... decimal places" or "significant figures".

Example 3.1 Solve $x^2 - 7x + 3 = 0$, giving your answer correct to two decimal places.

SOLUTION

$a = 1, b = -7, c = 3$; $b^2 - 4ac = 49 - 12 = 37$ and $\sqrt{37} \approx 6.083$

$x = \frac{7 \pm 6.083}{2}$, so $x = \frac{13.083}{2} = 6.5415$ or $x = \frac{0.917}{2} = 0.4585$

$x = 6.54$ **or** $x = 0.46$

REMEMBER

For an answer correct to 2 decimal places, work to 3 places and round off only at the end. For significant figures, count from the first non-zero digit: 0.4585 correct to 2 significant figures is 0.46.

4 Nature of the roots

- The **discriminant** is $D = b^2 - 4ac$.

Discriminant	Nature of the roots
$D > 0$	Real and unequal
$D = 0$	Real and equal, each $= -\frac{b}{2a}$
$D < 0$	Not real (imaginary)

- If D is a perfect square (and a, b, c are rational), the roots are rational; otherwise they are irrational.
- The roots are **real** when $D \geq 0$.

Example 4.1 Without solving, find the nature of the roots of (i) $3x^2 - 5x + 2 = 0$ (ii) $x^2 + 4x + 5 = 0$.

SOLUTION

(i) $D = 25 - 24 = 1 > 0$

(ii) $D = 16 - 20 = -4 < 0$

(i) Real and unequal (also rational, since 1 is a perfect square). (ii) Not real.

Example 4.2 Find the values of m for which $x^2 - 2mx + 9 = 0$ has equal roots.

SOLUTION

For equal roots, $D = 0$: $(-2m)^2 - 4(1)(9) = 0$

$4m^2 = 36$, so $m^2 = 9$

$m = 3$ **or** $m = -3$

5 Common mistakes

- Dividing both sides by x and losing the root $x = 0$.
- Reading b or c without its sign.
- Dividing only $\sqrt{D}$ by $2a$ instead of the whole numerator.
- Rounding too early, which makes the final decimal places wrong.
- Not checking that a solution keeps every denominator non-zero.